Conjecture 00000000642: the LKB determinant changes under stabilization

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1 The necessary determinant condition

The source says the LKB image is "Markov-invariant at the determinant level." The ordinary, unnormalized determinant already fails this necessary condition. Work over the actual rational function field $K = \mathbf{Q}(q,t)$, with q, t independent indeterminates. Under positive stabilization from two to three strands, the identity braid becomes σ_2. Their representation determinants are respectively

$$1 \qquad \text{and} \qquad tq^3 \neq 1.$$

Thus the stated conjunction is false under its literal determinant condition. No additional writhe correction or normalization is specified in the source. We make no claim about what an extra normalization could achieve, and do not separately settle its Jones-value or kernel assertions.

2 The standard fraction-field matrix model

Bigelow, *The Lawrence–Krammer representation*, Section 4.2, p.13, gives the Krammer generator action on the basis F_{ij}. He explicitly states that over $\mathbf{Q}(q,t)$ this model and the homological Lawrence–Krammer representation are isomorphic, hence conjugate as matrix representations. He also warns that their integral lattices need not be isomorphic. We stay over the generic fraction field throughout; we do not specialize the representation at numerical parameters. We use Bigelow's sign convention for t.

In the ordered basis (F_{12}, F_{13}, F_{23}), his displayed action gives

$$A = \begin{pmatrix} -tq^2 & 0 & q^2-q \\ 0 & 0 & q \\ 0 & 1 & 1-q \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & q & 0 \\ 1 & 1-q & 0 \\ 0 & (1-q)qt & -tq^2 \end{pmatrix}.$$

For two strands, the single generator acts on F_{12} by the 1×1 matrix $(-tq^2)$. Direct calculations, all checked in Lean, give

$$\det A = \det B = tq^3, \qquad ABA = BAB, \qquad \det(-tq^2) = -tq^2.$$

Since $q, t \neq 0$ in K, these matrices are invertible. Lean uses the genuine Artin presentations B_2 (one free generator) and $B_3 = \langle \sigma_1, \sigma_2 \mid \sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2 \rangle$. It proves the relation before constructing actual homomorphisms $\rho_2 : B_2 \to \mathrm{GL}_1(K)$ and $\rho_3 : B_3 \to \mathrm{GL}_3(K)$ by their universal properties.

3 A genuine elementary Markov move

The inclusion $\iota : B_2 \to B_3$ adds a straight third strand, sending the first generator to σ_1. Positive stabilization is

$$b \longmapsto \iota(b)\sigma_2.$$

This is the standard elementary Markov move: see Bigelow, Ramos and Yi, *The Alexander and Jones polynomials through representations of Rook algebras*, printed p.1250114-4, moves (2)–(3) preceding Theorem 1.5. Lean constructs this inclusion as a group homomorphism, defines the positive move by this very product, and proves that 1 stabilizes to σ_2. Consequently,

$$\det \rho_2(1) = 1, \qquad \det \rho_3(\iota(1)\sigma_2) = \det B = tq^3.$$

The closed theorem `not_positive_markov_invariant` negates invariance over all pairs related by this actual elementary move. Failure of one allowed move suffices; no full topological Markov theorem or construction of link closures is needed for this necessary condition.

4 Independent variables and arbitrary bases

The formal field is exactly

$$K = \mathrm{Frac}(\mathbf{Q}[X_0, X_1]), \qquad q = X_0, \quad t = X_1.$$

Injectivity into the fraction field proves $q, t \neq 0$. If $tq^3 = 1$ in K, injectivity would imply the polynomial identity $X_1 X_0^3 = 1$. Applying the polynomial evaluation homomorphism $X_0 = X_1 = 0$ gives $0 = 1$, a contradiction. This evaluation is applied only to an alleged polynomial identity. It is not a specialization of an LKB representation, and no evaluation on the whole fraction field is asserted.

For every invertible matrix P, Lean proves

$$\det(PMP^{-1}) = \det(M).$$

The theorem `determinant_changes_in_any_bases` therefore gives the same unequal determinants after arbitrary and independent changes of basis in dimensions one and three. This also explains why the generic conjugate LKB model has the same obstruction. The report uses the standard matrix model as defined by the primary source; Lean does not reconstruct configuration-space homology or reprove Bigelow's topological identification theorem.

5 Verification and scope

Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d` are pinned. The package contains a new root build, nineteen transitive axiom audits, and full official `leanchecker --fresh Main` replay, with source-bound commands, exit codes, logs and hashes. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed the formal chain and report. This is AI review, not human certification. Organizer acceptance is pending.